

① Dynamic Programming :- It is used to solve 2 type of Problem:-

1. Finding optimal solution for any problem (min-max values)

2. Count no of way in which problem could be done.

① Any question related to D.P can be solved via 2 ways:-

1. Top-down Approach

(Recursion + Memorization)

2. Bottom-Up Approach

(Here use 1-D or 2-D aux-array and start to fill from index 0 or (0,0) till size .

$$① \quad f(n) = f(n-1) + f(n-2)$$

$$f(\underline{0}) = 1, \quad f(\underline{1}) = 1$$

$$f(\underline{2}) = f(1) + f(0) = \underline{2}$$

$$f(3) = \underline{f(2)} + f(1) = 3$$

$$f(n) = \underline{f(3)} + \underline{f(2)} = 5$$